

NAME-KOTTA THARITH REDDY

SRN-PES1UG24CS233

CLASS-D

TOPIC-Multi-Cloud Asset Cost Optimization Portal.

DELIVERABLE 1

Req ID	Type	Description
FR-001	Functional	The system will take cloud billings as input to the CSV exports, categorize by 1. Cost of services . 2. project tags and highlight computer instance with <5% CPU utilization in the last 14 days.
FR-002	Functional	The system shall find unattached storage volumes across AWS, Azure, and GCP and show their estimated monthly cost
FR-003	Functional	The system shall track cloud spending against project or team budgets and send an alert when the actual or expected spending goes over the set budget.
FR-004	Functional	The system shall create a cost optimization report for each stakeholder showing idle assets, unattached storage, and possible monthly savings across AWS, Azure, and GCP.
FR-005	Functional	The system shall allow users to filter the cost dashboard by cloud provider, project tag, region, and date range. Only Cloud Administrators should be able to change settings such as alert thresholds.

NFR-001	Nonfunctional	The portal should process large cloud billing files and update the cost dashboard within 2 minutes.
NFR-002	Nonfunctional	The portal should be available 99.9% of the time each month and keep billing data encrypted while it is stored and being transferred.

SE LAB-1

Priority	Acceptance Criteria
High	Idle cloud assets should be identified with their estimated monthly savings. Active instances should not be marked as idle.
High	The system should detect an unattached volume within 24 hours and include it in the report with its estimated cost. If an unattached volume is older than 24 hours, it should not be missing from the report
High	The alert should be sent within 15 minutes after a budget breach is detected and should show the correct project, budget limit, and current spending. If the alert is not sent or is delayed by more than 15 minutes, it should be considered a failure.
Medium	The total in the report should match the total cost of all flagged assets for the same period. If the totals do not match, it should be considered a failure.
Medium	The filtered results should only show the matching records, and users should not be able to access settings they are not allowed to change. If unrelated records appear or an unauthorized user can change restricted settings, it should be considered a failure.

High	Testing should confirm that the portal meets the required speed and security standards during heavy usage.
High	The portal should be available at least 99.9% of the time over 30 days, and billing data should be encrypted when stored and transferred. If the uptime falls below 99.9% or any billing data is found to be unencrypted, it should be considered

Rationale
Idle compute can lead to a lot of unnecessary cloud spending, so finding it is an important part of the FinOps analyst's work.
Unattached storage volumes can continue to cost money even when they are not being used, so finding them is an easy way for cloud administrators to reduce costs.
Sending budget alerts quickly helps FinOps analysts take action before the extra spending gets too high.
Stakeholders need a single reconciled view across three providers rather than manually combining three separate billing exports.
There is a lot of data across multiple cloud providers, so filters are needed to find the required information quickly. Also, FinOps Analysts and Cloud Administrators should have different permissions for changing settings.

FinOps analysts need to see updated data quickly, especially during month-end work, so slow data processing would reduce the usefulness of the portal.

Billing data contains sensitive information about costs and infrastructure, so the portal needs to keep the data secure and also be available when users need it.